Day 4 morning — Two unknowns

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Friday 13 November 2026 · Kroghstræde 3, room 1.104 · group rooms 4.129 and 5.149

The normal model with μ and σ · joint, marginal and conditional posteriors · marginalising with draws · an average versus one attempt

Today

08:45	**Lecture A**	Two unknowns: joint, marginal, conditional · one new attempt
09:45	**Exercise A**	Normal model in Stan · joint draws · average vs attempt · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Grouped data · pooling · shrinkage · hierarchical models in Stan
13:30	**Exercise B**	Three pooling models · shrinkage · known vs new person · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

Follow-up from yesterday

From the Day 3 diaries and plenary

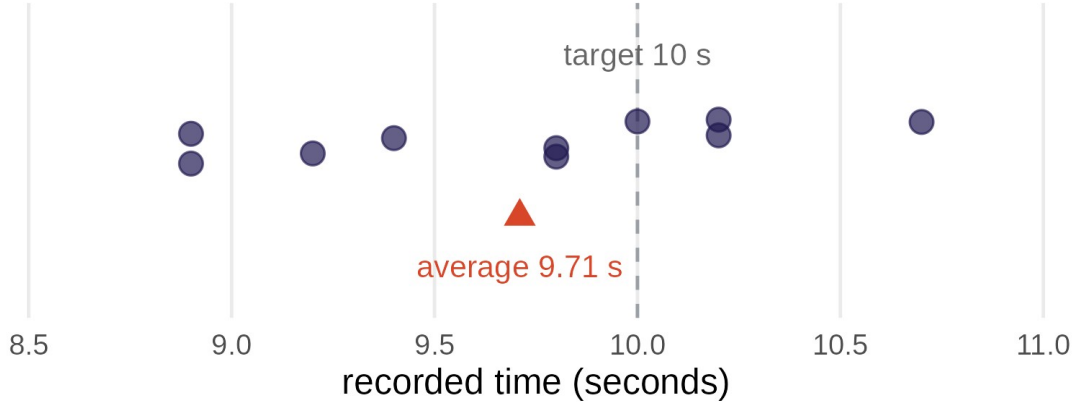
Fill in on Thursday evening: the two or three points that came up most often under “not yet familiar”.

1. …
2. …
3. …

By the end of this lecture you can

1. **Write** a model with two unknown parameters, the normal model with μ and σ
2. **Explain** joint, marginal and conditional posteriors
3. **Marginalise** with draws: use one column and ignore the others
4. **Fit** the model in Stan and read the draws as matched pairs
5. **Separate** uncertainty about an average from the prediction of one attempt

One person, ten attempts



P04 tried to wait 10 seconds, ten times, with eyes closed and no feedback (the Day 1 time game; illustrative numbers).

Three questions

1. What is her typical time?
2. How much do her attempts vary?
3. Will her next attempt be shorter than 10 s?

Questions 1 and 2 are about two different unknowns — and question 3 needs both.

A model with two unknowns

$$y_i \mid \mu, \sigma \sim \text{Normal}(\mu, \sigma), \quad i = 1, \ldots, 10, \quad \text{independent}$$

μ (mu)

Her long-run average time, in seconds.

Prior $\mu \sim \text{Normal}(10, 2)$: probably between 6 and 14 s.

σ (sigma)

How much her attempts spread around μ, in seconds.

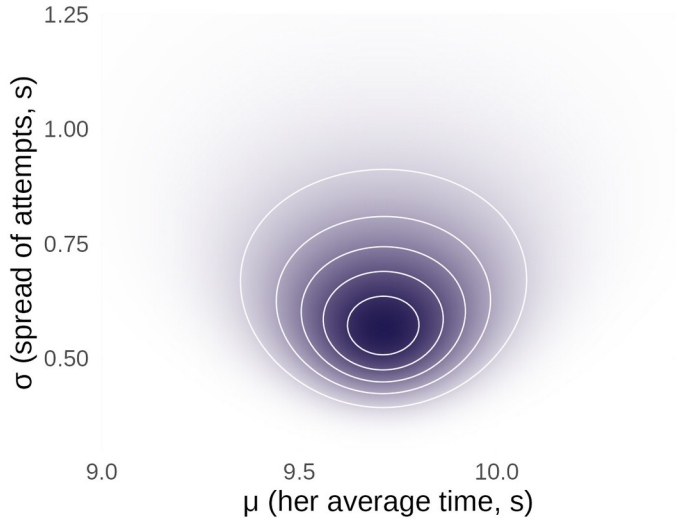
Prior $\sigma \sim \text{Normal}^{+}(0, 2)$ (half-normal, since $\sigma > 0$): the spread is probably below 4 s.

- Day 2 had one unknown (θ or λ). Now we learn two at once.
- Both are about this one person; the class comes this afternoon.

Bayes' rule with two parameters

$$p(\mu, \sigma \mid y) \propto p(y \mid \mu, \sigma) \cdot p(\mu) \cdot p(\sigma)$$

- Likelihood: $p(y \mid \mu, \sigma)$ = product of ten Normal($y_i \mid \mu, \sigma$) densities.
- The posterior is a surface over the (μ, σ) plane: one height for every pair of values.
- Here computed on a grid of 221 × 141 points — Day 2's grid, now in two dimensions.

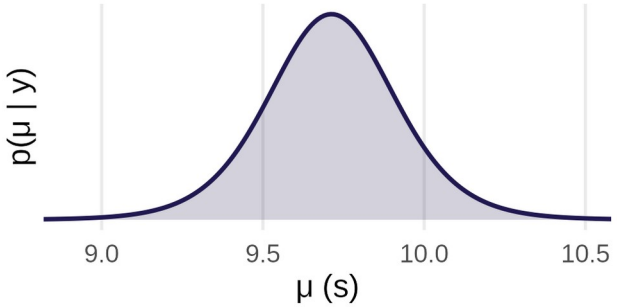


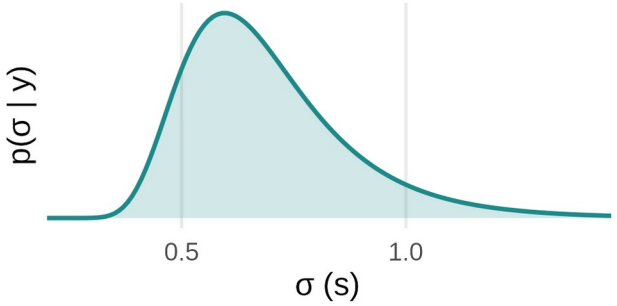
Darker = higher posterior density

Same recipe as before: likelihood × prior, then rescale. Only the picture has one more dimension.

Joint, marginal and conditional

posterior	question it answers	on the surface
joint $p(\mu, \sigma \mid y)$	which pairs (μ, σ) are plausible together?	the whole surface
marginal $p(\mu \mid y)$	what do we know about μ, whatever σ is?	add the surface up over σ
conditional $p(\mu \mid \sigma, y)$	what would we know about μ if σ were known?	one slice at a fixed σ





Usually we report marginals: one parameter at a time, with the uncertainty about the others included.

Marginalising on a grid

μ \ σ	0.5	0.7	0.9	p(μ | y)
9.5	0.08	0.10	0.06	**0.24**
9.7	0.14	0.18	0.10	**0.42**
9.9	0.10	0.14	0.10	**0.34**
p(σ | y)	**0.32**	**0.42**	**0.26**	1

A toy joint posterior on a 3 × 3 grid: the nine numbers add up to 1.

Marginal of μ: row sums

$p(\mu \mid y) = \int p(\mu, \sigma \mid y)\, d\sigma \approx \Sigma$ over σ

9.7: 0.14 + 0.18 + 0.10 = 0.42

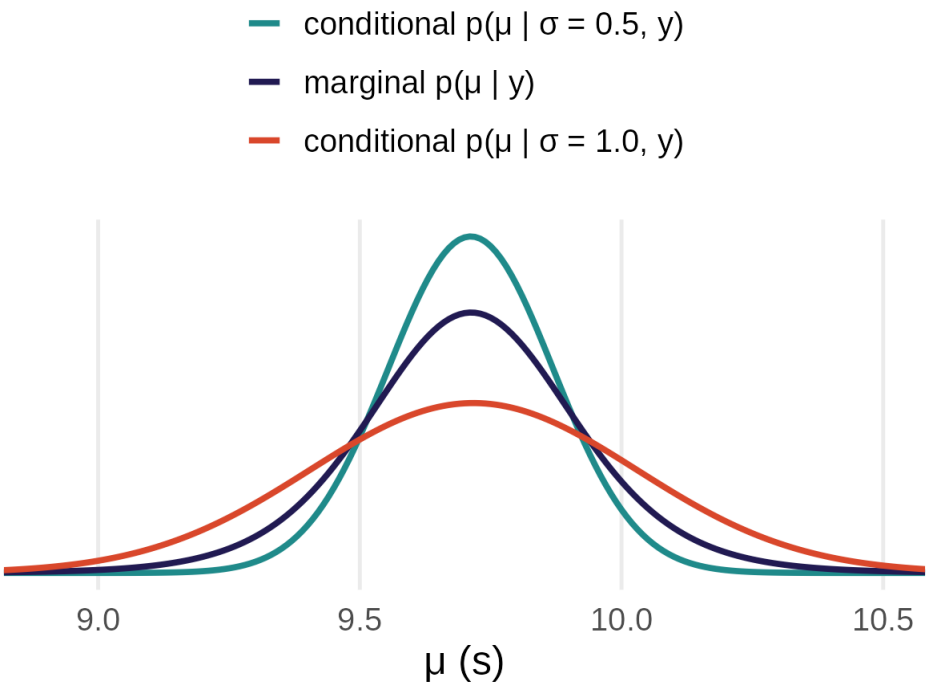
Conditional at σ = 0.5: one column, rescaled

0.08, 0.14, 0.10 divided by their sum 0.32

→ 0.25, 0.44, 0.31

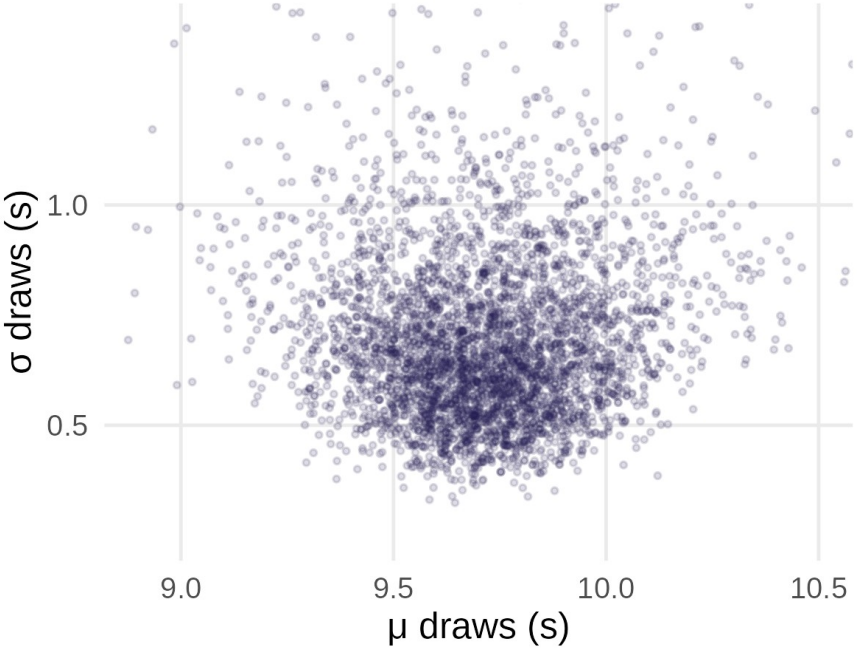
Marginal: add up over σ. Conditional: keep one value of σ and rescale.

Conditional is not marginal



- Fixing σ pretends that we know it.
- If σ were 0.5, we would be more certain about μ; if 1.0, less.
- The marginal averages over all values of σ, weighted by how plausible they are.
- Plugging in one value of σ hides our uncertainty about σ — like Day 2's plug-in prediction.

With draws, marginalising is free



draw	μ	σ
1	9.87	0.52
2	9.95	0.56
3	9.74	0.83
4	9.80	0.77
…	…	…

4,000 draws (Stan, next slides)

- Each Stan draw is a pair $(\mu^{(s)}, \sigma^{(s)})$ from the joint posterior.
- The μ column alone is a sample from the marginal $p(\mu \mid y)$: just ignore σ.
- No integral: this works the same with 100 parameters.

The normal model in Stan

```
data {
  int<lower=1> N;          // number of attempts
  vector[N] y;             // recorded times (s)
}
parameters {
  real mu;                 // her average time
  real<lower=0> sigma;     // spread of attempts
}
model {
  mu ~ normal(10, 2);      // prior
  sigma ~ normal(0, 2);    // half-normal: sigma > 0
  y ~ normal(mu, sigma);   // likelihood, all N at once
}
generated quantities {
  // her next attempt
  real y_new = normal_rng(mu, sigma);
}
```

- Two parameters, declared together.
- <lower=0> plus normal(0, 2) gives the half-normal prior.
- y ~ normal(mu, sigma) is the product of all N likelihood terms.
- y_new uses the μ and σ of the same draw.
  - Stan's normal(mu, sigma) uses the SD, like R's dnorm().

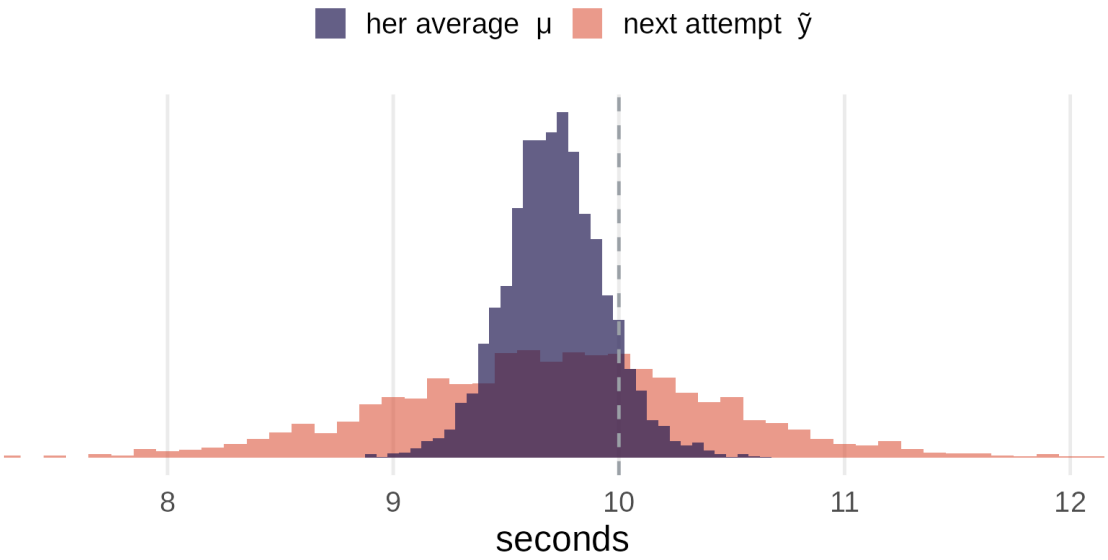
The fit: P04's posterior

```
fit <- mod$sample(data = list(N = 10, y = y), chains = 4, seed = 2026)
fit$summary(c("mu", "sigma", "y_new"))
```

variable	mean	sd	5%	95%	rhat	ess_bulk
mu	9.72	0.23	9.35	10.09	1.00	2039
sigma	0.70	0.19	0.45	1.06	1.00	2351
y_new	9.71	0.76	8.48	10.89	1.00	3867

- No divergences, no max-treedepth hits, E-BFMI fine: we may interpret.
- The grid gives the same means: μ 9.71, σ 0.69.

Her average versus her next attempt



Two steps, one draw at a time

1. take a draw $(\mu^{(s)}, \sigma^{(s)})$
2. $\tilde{y}^{(s)} \sim \text{Normal}(\mu^{(s)}, \sigma^{(s)})$

90% interval	
her average μ	9.35 – 10.09
next attempt ỹ	8.48 – 10.89

An average is not one more attempt: the prediction adds her attempt-to-attempt spread σ.

Questions the draws answer

question	from the draws	answer
Is she, on average, too fast?	mean(mu < 10)	$P(\mu < 10 \mid y) = 0.90$
Will her next attempt be too fast?	mean(y_new < 10)	$P(\tilde{y} < 10 \mid y) = 0.66$
Are her attempts erratic?	mean(sigma > 0.8)	$P(\sigma > 0.8 \mid y) = 0.24$
Next attempt within 0.5 s of 10?	mean(abs(y_new - 10) < 0.5)	0.50

Any question about μ, σ or a future attempt: write it as TRUE/FALSE per draw, then take the mean.

Worked example: average or attempt?

Question 1

$P(\mu < 10 \mid y) = 0.90$, but $P(\tilde{y} < 10 \mid y) = 0.66$.

Why are the two probabilities so different?

Question 2

Suppose P04 had made 100 attempts, with the same average and spread.

What happens to the 90% interval for μ, and to the one for her next attempt?

2 minutes with your neighbour.

Worked example: answers

Question 1

μ is her average: we are fairly sure it is below 10.

One attempt scatters around μ with SD σ ≈ 0.7 s, so a single attempt can easily land above 10 even if her average is below.

90% interval	10 attempts	100 attempts
μ	9.35 – 10.09	9.61 – 9.81
next attempt	8.48 – 10.89	8.70 – 10.77

μ: about 1/√n narrower. Next attempt: only a little narrower; it approaches μ ± 1.64 σ, never zero width.

More data shrink our uncertainty about parameters, not the variation of single attempts.

Exercise A · 09:45 – 11:10

- Fit the supplied normal model to one person's attempts from the time game.
- Plot the joint draws of μ and σ, and both marginals.
- A 90% interval for μ versus one for the next attempt.
- Probabilities from the draws: about μ, about σ, about a new attempt.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

11:10

peer review A

Take a break when it suits your group. Rotate roles: coder · checker · question writers · interviewer · reporter.

Peer review A · 11:10 – 11:55

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

What to take into lunch and Lecture B.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

```
sample(not_yet_drawn, 2)   # plenary draw
```